Proposal: AI powered experiments on $SL_2(\mathbb{Z}_p)$

Benjamin Krawciw, Cody Strouse, Jake Quinn, Wenzhuo Wu

January 25, 2026

We propose using AI assisted programming to explore the diameter of the group $SL_2(\mathbb{Z}_p)$. Linear groups are ubiquitous in physics and mathematics. $SL_2(\mathbb{Z}_p)$ in particular is key in the study of representation theory, finite groups, and number theory.

$SL_2(\mathbb{Z}_p)$ has two generators, S and T. Given any γ in $SL_2(\mathbb{Z}_p)$ we can consider how many applications of S and T are needed to reach it. There is a smallest such number, call it $\ell(\gamma)$, and as $SL_2(\mathbb{Z}_p)$ is finite, there exists some γ in the group such that $\ell(\gamma)$ is greater than or equal to any other $\ell(\alpha), \alpha \in SL_2(\mathbb{Z}_p)$. This value is known as the diameter of the group, or informally, God's number. Denote God's number by $G(SL_2(\mathbb{Z}_p))$.

It is known that $G(SL_2(\mathbb{Z}_p)) = O(\log(p))$, but the asymptotic constant C is not known. Also, God's numbers for $SL_2(\mathbb{Z}_p)$ are not usually computed or tabulated. By computing $G(SL_2(\mathbb{Z}_p))$ for many p values, we can find an experimental estimate for C.

The Cayley graphs of $SL_2(\mathbb{Z}_p)$ are defined with the group elements as the vertices, and the edges are formed by applying S or T to reach another vertex. This means that the edges are directed and colored (by the choice of S or T). These will be Ramanujan graphs, which are known for their incredibly rapid growth. By computing $G(SL_2(\mathbb{Z}_p))$, we are finding the radius of such graphs.

References